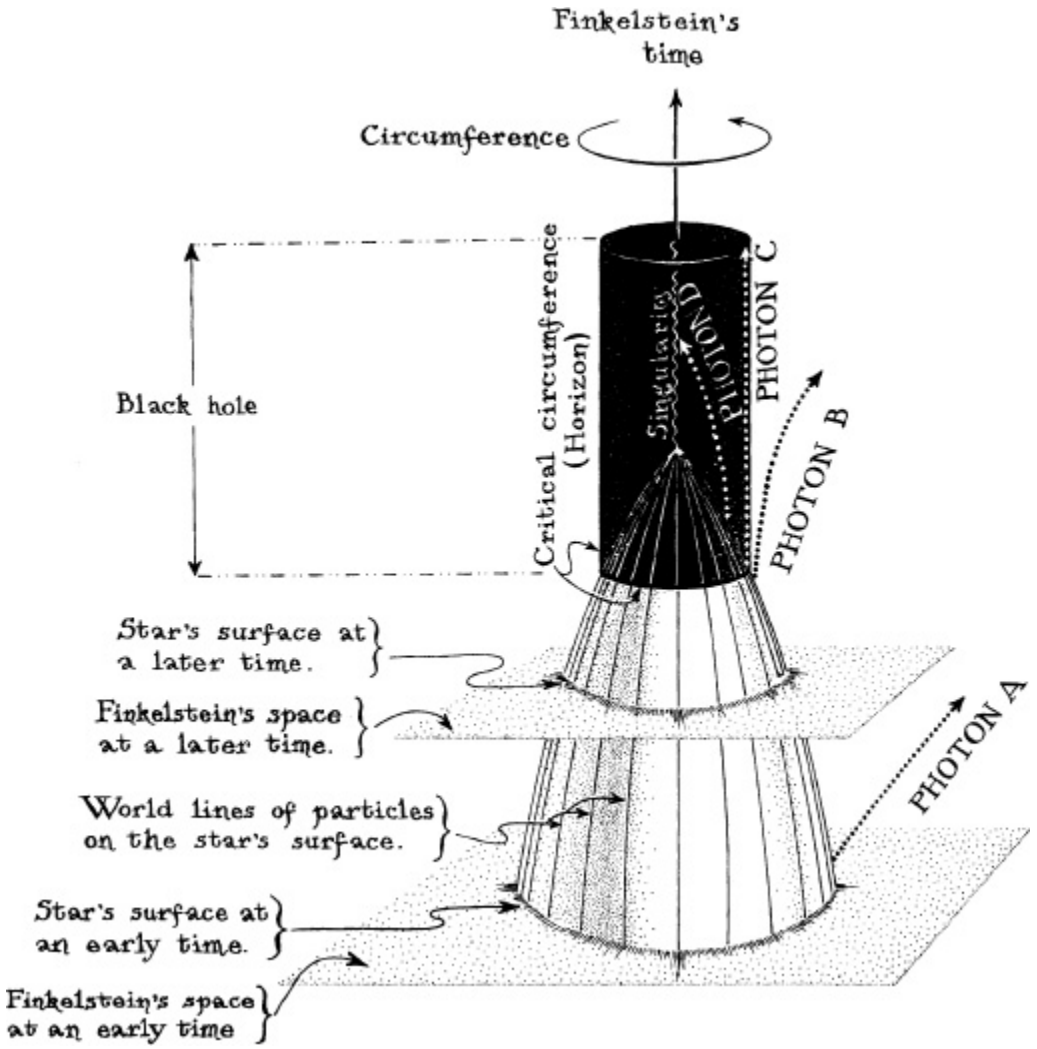


6.7 A spacetime diagram depicting the implosion of a star to form a black hole. Plotted upward is time as measured in Finkelstein's reference frame. Plotted horizontally are two of the three dimensions of that frame's space. Horizontal slices are two-dimensional "snapshots" of the imploding star and the black hole it creates at specific moments of Finkelstein's time, but with the curvature of space suppressed.

It is interesting to contrast this modern understanding of the propagation of light from an imploding star with eighteenth-century predictions for light emitted from a star smaller than its critical circumference.

Recall (Chapter 3) that in the late eighteenth century John Michell in England and Pierre Simon Laplace in France used Newton's laws of gravity and Newton's corpuscular description of light to predict the existence of black holes. These "Newtonian black holes" were actually static stars with circumferences so small (less than the critical circumference) that gravity prevented light from escaping from the stars' vicinities.